MICAH BAXTER

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EDUCATION

Bachelor of Science in Mechanical Engineering Expected May 2029

Liberty University, Lynchburg, VA

- Physics Minor, 3.85 GPA
- Organizations: Liberty Rocketry Competition Team – Chief Engineer 2026-2027
- Full Academic Scholarship, ACT: 35

TECHNICAL SKILLS

Software & Programming: SolidWorks, Onshape, Altium, KiCad, C#, C/C++, Python

Machines: Bambu Labs P1S combo (Bambu Studio), Creality Ender 3 (Cura), Shop Tools

EXPERIENCE

Information Technology Intern Summer 2026

Medical Properties Trust, Birmingham, AL

- Resolved hardware and software issues including printer connectivity, email configuration, and network problems in a corporate healthcare environment
- Configured new employee workstations for onboarding, ensuring systems met security and access standards
- Assisted IT administrators in managing Active Directory including directory searches and file access permissions for employees

Avionics Design Engineer 2025-2026

Liberty Rocketry Competition Team, Lynchburg, VA

- Developed a custom STM32-based flight computer to control critical sensor and telemetry operations for a high-power competition rocket
- Appointed as Liberty Rocketry Chief Engineer for the upcoming competition season

Lead Mentor Summer 2025

Tennessee Robotics Center, Nashville, TN

- Primary instructor and developer of TRC classes: 3D Printing and Engineering Design, Electronics and Robotics with Arduino, and REV Robotics

PERSONAL PROJECTS

Thrust Vector Control Rocket 2023-2025

- Self-taught CAD design, PCB design, and computer programming
- Engineered an autonomous launchpad with launch clamps, a strongback, and flame trenches
- Designed and flew two TVC rockets across different airframe scales with a custom PCB flight computer and deployable catch points to enable future catch attempts

Embedded Systems and PCB Design 2023-2026

- Interfaced the DPS310 barometric pressure sensor and the BNO055 absolute orientation sensor with a Teensy 4.1 microcontroller to stabilize the TVC rocket and trigger events based on altitude
- Developed an STM32-based microcontroller ground computer for an upgraded launchpad that implements I2C, SPI, and UART communication and includes PWM controlled GPIO headers for modularity
- Designed and manufactured the custom STM32-based flight computer for a high-power collegiate rocket that analyzes sensor and GPS data and manages live telemetry

Puzzle Boxes 2025

- Invented and branded Baxter Boxes, a unique line of custom 3D printed puzzle boxes including HEX and a special Easter egg box with intricate details that function as mechanical moving parts